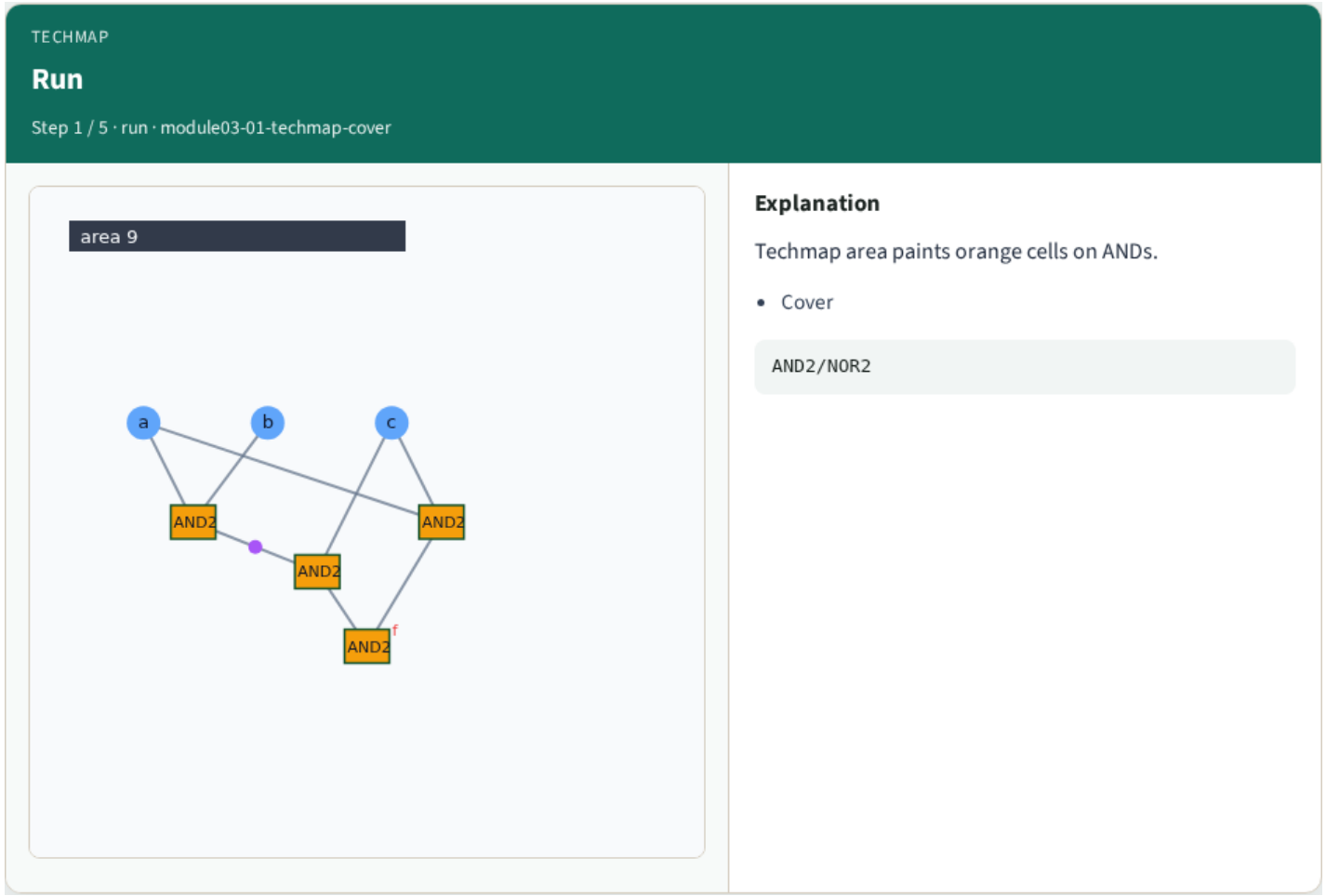
Techmap

A cover assigns a library cell to each subject node (or cut)

The idea

- Greedy: AND→AND2, AND(~,~)→NOR2, plus INV per complemented edge; sum areas

Run


TECHMAP
Run
Step 1 / 5 · run · module03-01-techmap-cover
area 9
a
b
c
AND2
AND2
AND2
AND2
f
Explanation
Techmap area paints orange cells on ANDs.
Cover
AND2/NOR2

Run

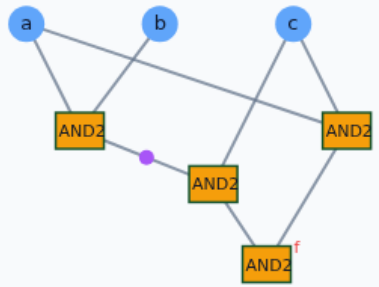
Area

TECHMAP

Area

Step 2 / 5 · area · module03-01-techmap-cover

area 9



Explanation

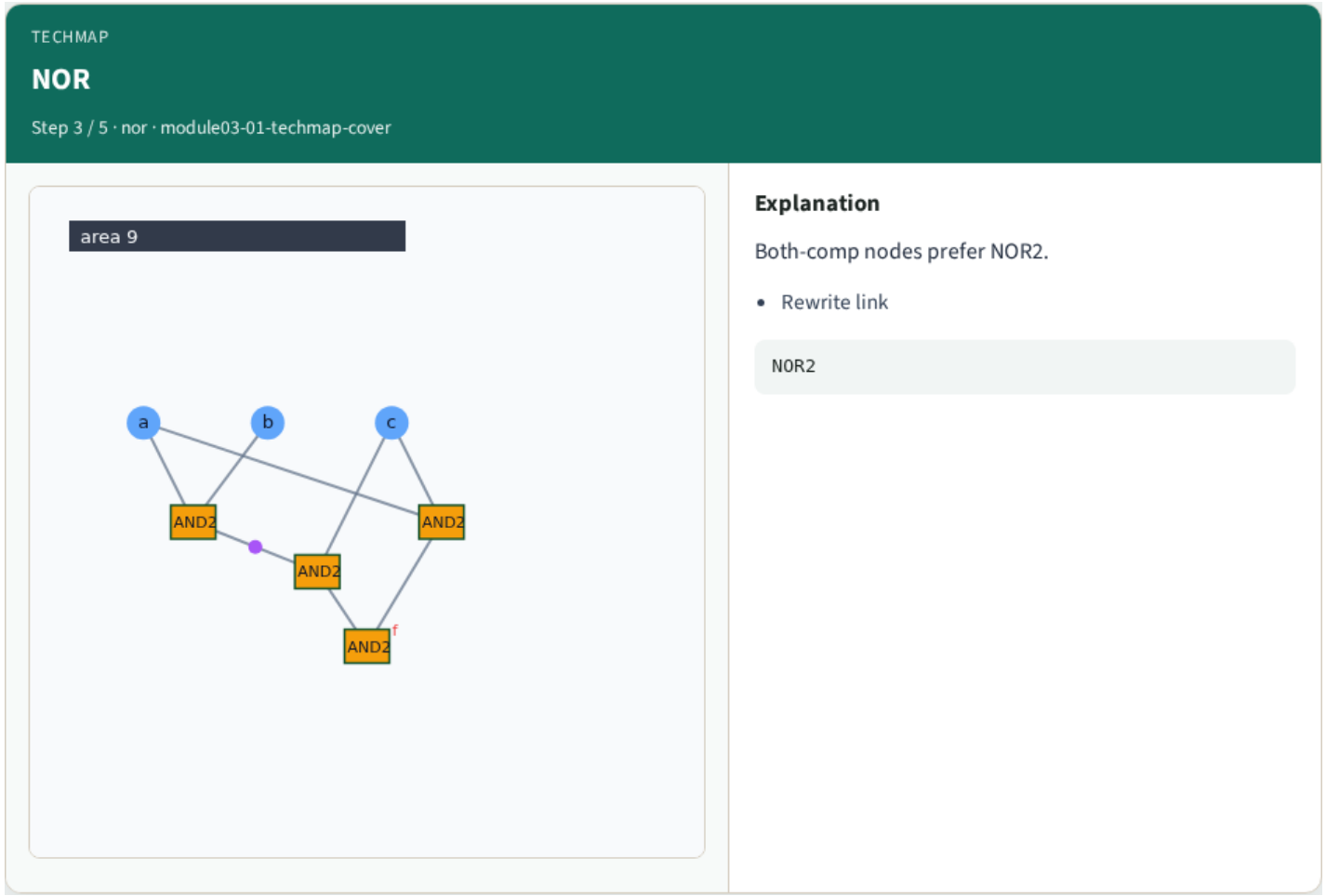
Total area includes AND cells plus INVs.

- Sum

≥8

Area

NOR


TECHMAP
NOR
Step 3 / 5 · nor · module03-01-techmap-cover
area 9
a
b
c
AND2
AND2
AND2
AND2
f
Explanation
Both-comp nodes prefer NOR2.
Rewrite link
NOR2

NOR

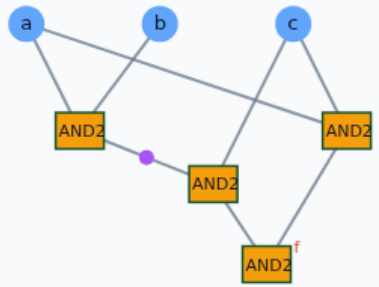
Inv

TECHMAP

Inv

Step 4 / 5 · inv · module03-01-techmap-cover

inv 1



Explanation

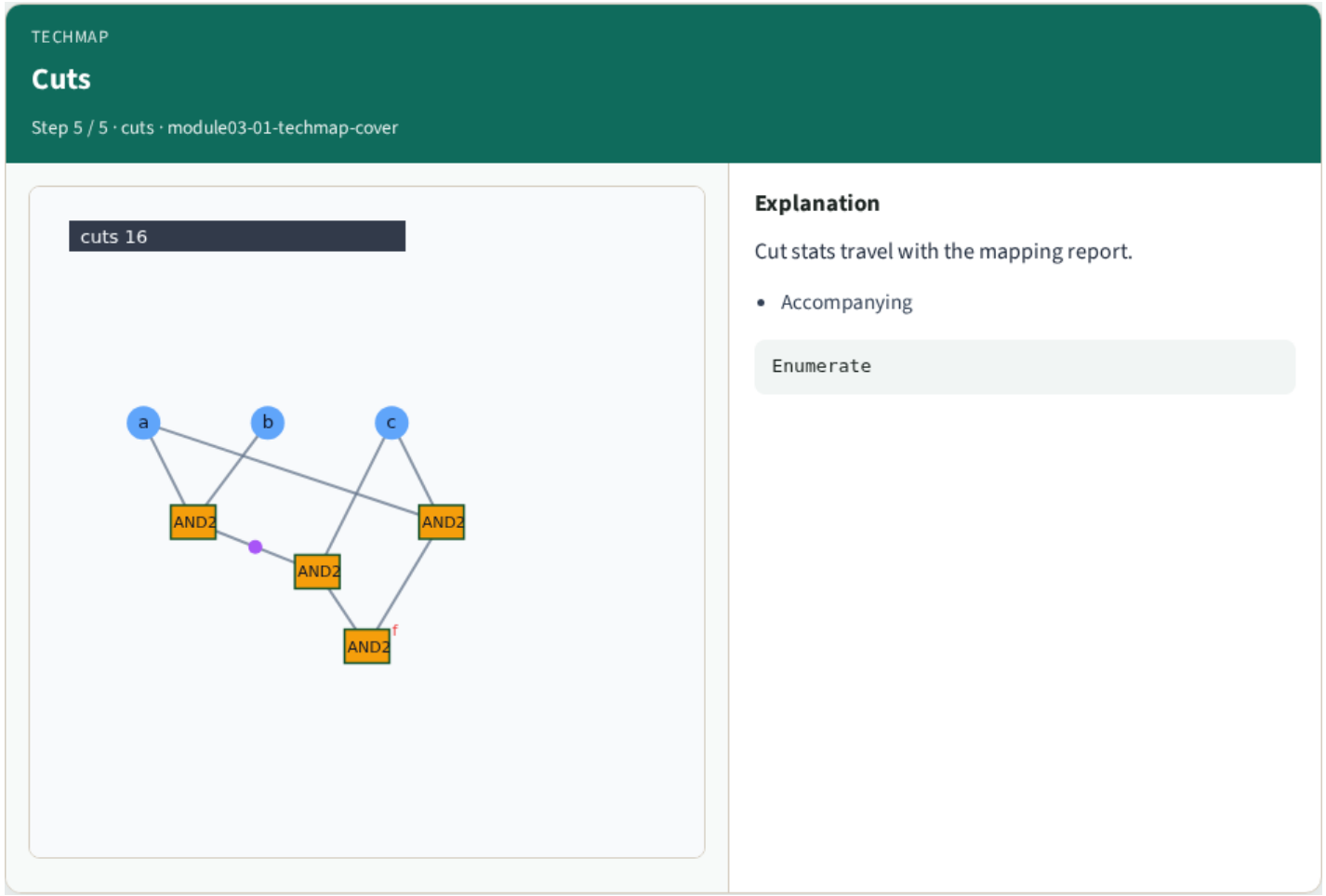
Complement bubbles add INV area.

- invArea

```
Cost
```

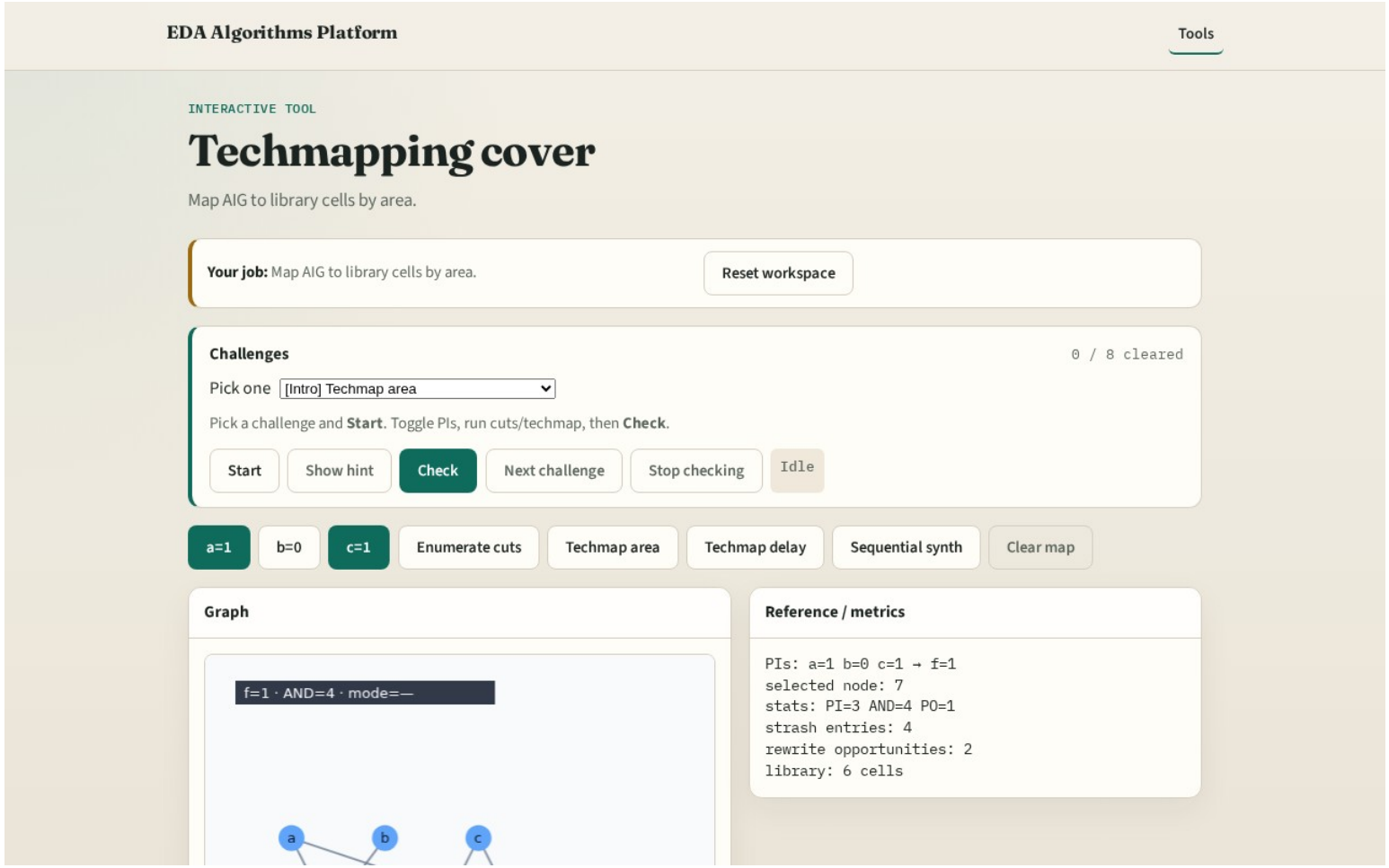
Inv

Cuts


TECHMAP
Cuts
Step 5 / 5 · cuts · module03-01-techmap-cover
cuts 16
a
b
c
AND2
AND2
AND2
AND2
f
Explanation
Cut stats travel with the mapping report.
Accompanying
Enumerate

Cuts

Browser lab track



Browser lab starter

Implement track

- Call techmap_area; assert area ≥ mapAreaMin

Pitfalls

- Forgetting INV cost
- Mapping PIs to gates

Your turn

- Next: area vs delay tradeoff